

Data Quality Profiling – Sprint 1

Project: Heavy Supplier, Inventory & Warehouse Analytics

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Sprint: Sprint 1 – Data Foundation & Exploration

1. Objective

The objective of this data-quality profiling is to assess the quality, consistency, completeness, and reliability of the available supply-chain datasets before performing further analysis.

The following checks were performed:

- Missing values
- Duplicate records
- Duplicate primary/composite keys
- Date consistency
- Unusual or invalid values
- Cross-table consistency
- Basic numerical and calculation checks

2. Dataset Scope

The analysis covered all 12 tables in the project dataset:

1. Branches
2. Customers
3. Inventory Master
4. Invoices
5. Payments
6. Products
7. Purchase Orders Header
8. Purchase Orders Lines
9. Sales Orders Header
10. Sales Orders Lines
11. Stock Ledger
12. Suppliers

3. Missing Value Analysis

Most columns across the datasets were complete with no missing values.

The only significant missing-value pattern was found in:

Table	Column	Missing Values
purchase_orders_header	received_date	2,370

Further investigation showed that there were exactly 2,370 cancelled purchase orders, matching the number of missing "received_date" values.

This indicates that the missing dates are likely expected/conditional missing values, because cancelled purchase orders would not necessarily have a received date.

Finding: No major random missing-data problem was identified.

4. Duplicate Analysis

Exact Duplicate Rows

No exact duplicate rows were found in any of the 12 tables.

Result: 0 exact duplicate records.

Key Duplicate Checks

Most primary and composite keys were unique. However, two important issues were identified:

Table| Key| Finding

invoices| invoice_id| 393 affected rows across 196 duplicate ID groups

payments| payment_id| 403 affected rows across 201 duplicate ID groups

These duplicate IDs should be investigated because they may cause incorrect aggregations, double counting, or problems when joining tables.

Severity: High – requires validation before downstream analysis.

5. Date Consistency Analysis

Date fields were checked for valid date formats and logical date relationships.

Positive Findings

The following checks showed no issues:

- Purchase order "order_date" was not after "expected_delivery_date".
- Purchase order "received_date" was not before "order_date".
- Sales order "order_date" was not after "delivery_date".
- Delivered sales orders did not have missing delivery dates.
- Invoice "invoice_date" was not after "due_date".
- Customer "last_purchase_date" was not before "customer_since".

Issues / Business Observations

Late Purchase Order Deliveries

A total of 3,267 purchase orders were received after their expected delivery date.

This is not necessarily invalid data, but it is an important operational issue that can be used to measure supplier performance and delivery delays.

Payment Date Before Invoice Date

There were 208 payment records where the payment date occurred before the invoice date.

These records should be investigated because they may indicate data-entry errors, incorrect dates, or unusual advance-payment situations.

Payments After Due Date

There were 8,200 payments made after the invoice due date.

This should primarily be treated as a late-payment indicator, rather than automatically marking the records as invalid.

6. Payment Status Consistency

There were 1,837 invoices marked as "Unpaid".

Among these, 35 invoices had an associated payment record.

This indicates a possible inconsistency between the invoice payment status and the payment transaction data.

Recommendation: These records should be reviewed to determine whether the invoice status needs to be updated or whether the payment records represent partial/incorrect transactions.

7. Unusual / Invalid Values

Numerical fields were checked for negative, zero, and logically unusual values.

No significant negative or zero-value issues were identified in the checked quantity, price, cost, and financial fields.

Additional checks showed:

- Product selling price was greater than product cost for all checked products.
- No negative product margins were identified.
- Purchase-order line calculations matched quantity × unit cost.
- Sales-order line calculations matched quantity × unit price.
- Invoice total calculations were consistent.
- Sales-order total calculations were consistent.

- Purchase-order total calculations were consistent.

Result: The major financial and quantity calculations appear internally consistent.

8. Referential Integrity

Foreign-key relationships were checked between the relevant tables.

No unmatched foreign-key records were identified.

This means that the expected relationships between customers, branches, products, suppliers, orders, invoices, payments, and stock records are structurally consistent.

Result: 0 unmatched foreign-key records in the checked relationships.

9. Key Data Quality Findings

Issue	Table / Column	Count	Assessment
Missing received date	purchase_orders_header.received_date	2,370	Likely expected for cancelled POs
Duplicate invoice IDs	invoices.invoice_id	393 rows / 196 groups	High priority
Duplicate payment IDs	payments.payment_id	403 rows / 201 groups	High priority
Late PO receipts	purchase_orders_header	3,267	Operational issue
Payment before invoice date	payments	208	Requires validation
Payment after due date	payments	8,200	Late-payment indicator
Unpaid invoices with payment records	invoices/payments	35	Status consistency issue

10. Overall Data Quality Assessment

Overall, the dataset is suitable for further analysis, but a few issues should be investigated before creating final KPIs and dashboards.

The most important concerns are the duplicate "invoice_id" and "payment_id" values because they can potentially lead to double counting during joins and aggregations.

The missing "received_date" values appear to be logically explained by cancelled purchase orders and therefore do not appear to represent random missing data.

Late purchase-order deliveries and late payments are useful business findings and can also become important KPIs for supplier and financial performance analysis.

11. Recommendations

1. Investigate duplicate "invoice_id" values before invoice-level aggregation.
2. Investigate duplicate "payment_id" values before payment analysis.
3. Validate the 208 payments that occur before their invoice dates.
4. Review the 35 invoices marked as unpaid despite having payment records.
5. Use late purchase-order deliveries as a supplier-performance KPI.

6. Use payment delays as a financial-performance KPI.
7. Keep the missing "received_date" values for cancelled POs rather than filling them artificially.
8. Document all data-quality issues before building the final dashboard/model.

12. Conclusion

The data-quality profiling shows that the dataset is generally structured and usable for analytics. There are no exact duplicate rows, major calculation mismatches, or broken foreign-key relationships.

The main areas requiring attention are duplicate invoice/payment IDs, payment-date inconsistencies, invoice-payment status mismatches, and late purchase-order deliveries.

After validating these issues, the dataset can be used confidently for defining business questions, KPIs, and further analytics in the next stages of the project.